

Supplementary material: Masking cannot remove what is inside the region of interest

S1. Pre-registration index

Every experiment was specified in a dated document committed to the public repository before any model was fitted on the corresponding data; amendments were committed before the amended analysis. Git history provides the time stamps.

Document	Scope
PREREGISTRATION_ISIC2019_TRAPS	dermoscopy hair traps (and the author's E13 specification)
PREREGISTRATION_THYROID_TRAPS	thyroid calipers: real traps, controlled sweeps (DINOv2, MedSigLIP)
PREREGISTRATION_OVARY_TRAPS	ovarian calipers: real traps, controlled sweep
PREREGISTRATION_CAPSULE_TRAPS	capsule debris: real traps, controlled sweeps
PREREGISTRATION_CXR_DEVICE_TRAPS	chest-radiograph devices (CLiP) and controlled tube sweep
PREREGISTRATION_CXR_DRAIN	real chest drains (in-ROI only)
PREREGISTRATION_UNIVERSAL_I2E	U-MtE, ablations U7–U14 (erase vs augment, protection, JTT, SPLINCE, selectors)
PREREGISTRATION_SLAS	patch-token artifact localisation and suppression
PREREGISTRATION_FINETUNE	end-to-end fine-tuning (ResNet-50, ViT-S)
PREREGISTRATION_INPAINT_LAMA	LaMa inpainting baseline
PREREGISTRATION_NATURAL	unaltered-distribution and leave-one-hospital-out tests

The primary family used for multiplicity control (`STATISTICAL_PLAN`) was assembled after the individual results were available; each member test was individually pre-registered, and the Holm correction is applied to the whole family.

S2. Replication of the original pilot

Of 88 numbers reported in the thesis proposal, 73 are reproduced exactly (same sign, same confidence-interval verdict, $|\Delta| \leq 0.02$), 12 agree in sign and significance with a larger difference, and 3 differ (a borderline occlusion interval, a contrast trap that was reported as null and is not, and one source stratum). Details: `results/SUMMARY.md`.

S3. Datasets considered and not used

CANDID-PTX (not accessible outside Health New Zealand); RANZCR-CLiP devices linked to NIH (masking helped at both locations; device placement confounded with other devices); chest-radiograph cardiomegaly with lines over the heart (210 labelled images); TCGA/GrandQC pathology (pen-mark masks unreliable); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (caliper-free group only $\approx 70\%$ pure after a two-detector design); DDTI thyroid (TI-RADS instead of pathology); BKAI-IGH colonoscopy specular highlights (no highlight-free images).

S4. Artifact-mask audit and sensitivity

Visual audit of 40 random images per cell (thyroid detections: 95% on real calipers; caliper-free thyroid images: 95–97% without visible calipers; ovary: 40/40 detections are overlays, 31/40 calipers; caliper-free ovary images: 92% without calipers but $\approx 28\%$ with other overlays). Capsule debris probe: pixel accuracy 0.92, IoU 0.62 against expert masks. Re-running with large-marker (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions (thyroid, ovary) and a strict capsule Trap B leaves every primary conclusion unchanged (SENSITIVITY_ARTIFACT_LABELS).

S5. Secondary method analyses

- *Patch-token suppression (SLAS)*. A linear probe on DINOv2 patch tokens localises real hair from 5 annotated images (patch AUROC 0.93; 0.97 with 50), but dropping flagged tokens improves reversed AUROC only modestly (+0.02 hair, +0.08 thyroid over ROI pooling).
- *Annotation-free pseudo-groups*. An overlay detector trained on synthetic insertion pairs recognises real calipers (AUROC 0.84) but not hair (0.57); pseudo-group balancing adds +0.04 on thyroid only.
- *Erasure combined with DFR*. Adds ≤ 0.014 over masking plus DFR.
- *Split-validation selector*. Admitting DFR as a candidate (trained and scored on disjoint validation halves) fixes the chest-drain case (0.656 vs 0.538) but makes choices noisier elsewhere; neither selector is uniformly best.

S6. Leakage

Holding cohort, features and protocol fixed, replacing the grouped split with random image-level splits (93 near-duplicate pairs across train/test) inflates the measured shortcut gap by 18% (0% overlap) and 10% (100%) and distorts both masking effects ($+0.184 \rightarrow +0.223$; $-0.129 \rightarrow -0.088$).

S7. Reproducibility

All numbers in the main paper and this supplement are produced by scripts in the repository (**scripts/**); tables and figures are generated, never typed. A two-command tool applies U-MtE and SLAS to new data with any frozen backbone; it reproduces the experimental features exactly (cosine similarity 1.0).